Learning Causal Representations Based on a GAE Embedded Autoencoder

Kuang Zhou, Ming Jiang, Bogdan Gabrys, *Senior Member, IEEE*, and Yong Xu

***Abstract*—Traditional machine-learning approaches face limitations when confronted with insufficient data. Transfer learning addresses this by leveraging knowledge from closely related domains. The key in transfer learning is to find a transferable feature representation to enhance cross-domain classification models. However, in some scenarios, some features correlated with samples in the source domain may not be relevant to those in the target. Causal inference enables us to uncover the underlying patterns and mechanisms within the data, mitigating the impact of confounding factors. Nevertheless, most existing causal inference algorithms have limitations when applied to high-dimensional datasets with nonlinear causal relationships. In this work, a new causal representation method based on a Graph autoencoder embedded AutoEncoder, named GeAE, is introduced to learn invariant representations across domains. The proposed approach employs a causal structure learning module, similar to a graph autoencoder, to account for nonlinear causal relationships present in the data. Moreover, the cross-entropy loss as well as the causal structure learning loss and the reconstruction loss are incorporated in the objective function designed in a united autoencoder. This method allows for the handling of high-dimensional data and can provide effective representations for cross-domain classification tasks. Experimental results on generated and real-world datasets demonstrate the effectiveness of GeAE compared with the state-of-the-art methods.**

***Index Terms*—Causal representation, causal discovery, autoencoder, transfer learning.**

I. Introduction

When it comes to developing an effective machine learning model, acquiring the requisite quantity and quality of data is crucial. Unfortunately, there are occasions when obtaining a sufficient amount of data within the target domain proves challenging. Transfer learning attempts to address the scenario by transferring knowledge from the source domain to train a reliable model that can be applied to a relevant target domain [1]. In recent years, it has been widely applied in many cross-domain tasks, including classification [2], target recognition [3], clustering [4] and fault diagnosis [5]. The key idea behind these works is to learn effective transferable representations. As a prominent type of transfer learning methodology, unsupervised domain adaptation (UDA) can leverage both labeled source domain data and unlabeled target domain data to acquire feature information for various tasks within the target domain [6], [7], [8], [9], [10], [11]. Alternatively, some approaches may demand multiple labeled datasets related to the target to collaboratively aid in the construction of features for the cross-domain prediction models, as observed in domain generalization (DG) [12], [13]. Generally, in UDA and DG, the feature representation process employs various techniques to describe the correlation between input variables, aiming to identify the most effective representation for the cross-domain classification tasks. However, in practice, some patterns correlated with the target in the training data may not be inherently relevant to the learning problem, such as the image backgrounds when classifying the foregrounds. Consequently, the potential spurious correlations between features and label variables can result in a decline in model performance and may even lead to negative transfer.

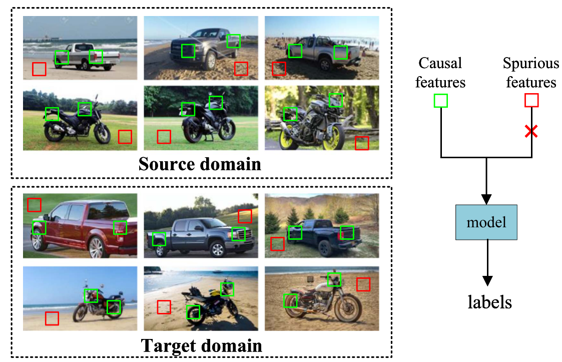

Fig. 1. Illustration of a classification task on a vehicle dataset in [14]. The features can be extracted from the image background, indicated by red boxes, and critical components, indicated by green boxes.

Causal relationships have the ability to capture the fundamental mechanism of data generation and are stable across various contexts. Taking the vehicle classification task illustrated in Fig. 1 as an example, we argue that features taken from these images can be roughly divided into causal features and spurious features. Causal features (highlighted by green boxes in the figure) play a crucial role in identifying the vehicle type (such as

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trucks, motorcycles, *etc*.). These features demonstrate stability and robustness across diverse environmental conditions. On the contrary, spurious features (such as certain environmental elements in image backgrounds, denoted by red boxes in the figure) can be considered confounding factors that may lead to distribution shift and cause performance degradation of predictive models on the unknown target domain.

In order to tackle these challenges in the feature representation process, some approaches strive to extract causal representations from observational data. In [15], [16], the authors employ causal interventions to disentangle causal factors from non-causal ones. However, these methods rely on strong assumptions. Furthermore, some score-based methods have been proposed to learn causal relationships between the features and target labels [17], [18]. Nevertheless, the complexity of these methods increases significantly with higher feature dimensionality.

Autoencoders, designed as a type of neural network model for learning representations, demonstrate the capability to map the input data into an effective lower-dimensional representation [19], [20]. Consequently, autoencoders have better representation capabilities for high-dimensional datasets and have already been applied in causal representation learning [21], [22]. Yang et al. proposed a causal autoencoder (CAE) model that combines a deep autoencoder with a causal structure learning frame [22]. The method can learn the causal representations by using data from a single labeled source domain effectively. The causal structure learning process in CAE is built on the assumption of a linear structural equation model (SEM), which limits its ability to capture nonlinear causal relationships between features and label variables. However, causal features have the potential to exhibit nonlinear relationships with the label variable. For example, the relationship between the vehicle type and its causal features (such as certain image features) is intuitively intricate and may not follow a linear pattern. Moreover, CAE simply learns the invariant representations through the alternating optimization of an autoencoder and a causal structure learning model, which decreases its scalability and efficiency.

In our work, a new causal representation method named GeAE via a graph-autoencoder-embedded autoencoder is proposed to address the aforementioned issues. GeAE is grounded in a nonlinear structural equation model formulated in a framework similar to a graph autoencoder (GAE). This approach enables the effective learning of causal relationships between feature variables and target labels. Furthermore, the causal structure learning loss, the cross-entropy loss as well as the reconstruction loss are well integrated into the autoencoder framework to learn the optimal causal representations solely from the source domain data for the cross-domain classification task effectively. Causal features that exhibit persistent relationships with the labels are utilized to construct the prediction model for the target domain, while spurious features are eliminated through the causal representation process. The main contributions of this study are summarized as follows:

- We have developed an efficient approach for learning causal representations in cross-domain classification. It integrates various losses (causal structure learning loss, cross-entropy loss, and reconstruction loss) within a unified framework using a GAE-embedded autoencoder. This enables the method to effectively learn nonlinear causal relationships between features and target labels, and makes it suitable for high-dimensional datasets.
- The proposed approach can provide domain-invariant representations, which in turn facilitate the construction of a predictive model that is applicable to unknown target domains. It effectively addresses the challenges of cross-domain classification where only a single labeled source domain is available.
- Experimental results conducted on synthetic datasets and four benchmark datasets demonstrate that GeAE can effectively identify robust causal features that significantly contribute to improving the learning performance on unseen target domains.

The remainder of this paper is organized as follows. Some related work is introduced in Section II. The proposed causal representation model is presented in detail in Section III. The experimental results are reported in Section IV. Conclusions are drawn in the final section.

II. Related Work

As mentioned before, the proposed GeAE model aims to address the challenges of feature representation for cross-domain classification by utilizing one single labeled source domain. This allows us to construct a predictive model capable of generalizing to any unseen target domain. The acquisition of invariant representations across domains through the feature extraction process is a pivotal concern. Therefore, this section will provide a comprehensive review of some related work including classical and causality-based feature selection methodologies and transfer learning techniques.

A. Classical Feature Selection

Feature selection, as one of the essential problems in machine learning, can eliminate irrelevant and redundant features from datasets. Classical feature selection methods typically depend on the correlation between features and target labels. Representative correlation-based feature selection methods include FCBF [23], mRMR [24], CFS [25] and JMIM [26]. FCBF is a fast filtering method that can identify relevant features and their redundancy. mRMR can maximize the correlation between feature subsets and categories while minimizing the redundancy among features. CFS can find strongly correlated non-redundant features by using the forward search strategy. JMIM considers the complementary information of features by calculating the correlation of category conditions. In real applications when there are some confounding factors, these approaches can lead to the inclusion of irrelevant features that are only connected to the specific contexts.

B. Causal Feature Selection

As mentioned, causal relationships remain stable across environments. Compared to correlation-based methods, causal features can eliminate the impact of spurious correlations and enhance robustness [27]. This is crucial for improving the performance of cross-domain prediction tasks in distribution drift scenarios. Generally, the objective of causal feature selection is to find the Markov Blanket (MB) or its subset of the label variable. MB can effectively capture the local causality of the label variable. When MB is utilized as a conditional set, features other than MB are conditionally independent of the label variable.

Recently, many causal feature selection methods have been proposed [17], [28], [29], [30]. These methods can be categorized into structure learning-based methods and intervention-based methods, depending on the techniques used to identify causal relationships.

For the first type, structure learning-based methods, Directed Acyclic Graphs (DAGs) are typically used to describe the causal relationships between variables [31], [32]. These methods can mainly be divided into constraint-based methods, score-based methods, and continuous optimization-based methods [33]. Constraint-based methods (e.g., BAMB [28] and DCMB [29]) use the conditional independence tests to discover the MB of the target labels, while score-based methods (e.g., [17]) use Bayesian score functions to discover the MB. The constraint-based methods cannot distinguish between the parent nodes and the child nodes of the label variable. Although score-based methods can identify this, they require learning the Bayesian network structure that involves all the currently selected features in each iteration. This can lead to a significant increase in computational complexity as the number of features increases. Therefore, these methods are not effective when applied to high dimensional datasets.

Recently, some optimization-based causal structure learning methods have been introduced, leading to significant improvements in computational efficiency [34], [35], [36]. Zheng et al. [34] proposed a method called NO TEARS, which formulates the structure learning problem as a purely continuous optimization problem over real matrices, thereby completely avoiding combinatorial constraints. Following this, Yang et al. [22] integrated a deep autoencoder and a causal structure learning model to learn causal representations using data. However, both of these methods rely on the assumption of a linear generation mechanism and are unable to effectively capture nonlinear causality. To address this issue, Yu et al. [35] introduced a polynomial-form acyclicity constraint for learning DAGs within the variational autoencoder framework, enabling the modeling of nonlinear causal relationships. Ng et al. [36] developed a network model using nonlinear structural equations to learn the adjacency matrix of a DAG within the graph autoencoder framework.

For the second type, intervention-based methods begin by estimating causal effects through causal interventions and other techniques, and then use this information to infer causal relationships between variables and identify causal features. Lv et al. [15] proposed three properties for causal factors and used a causal intervention module to generate domain-related perturbation data, separating causal factors from non-causal ones. Fu et al. [37] used causal interventions to identify causal effects, achieving domain-invariant representations by removing spurious associations.

Most of the aforementioned methods assume linear causal relationships, rendering them ineffective in identifying nonlinear causal structures. They also demonstrate low efficiency on high-dimensional data. Furthermore, they often lack designs for downstream classification tasks, compromising the effectiveness of representations. Thus, developing causal representation learning models for high-dimensional nonlinear data and integrating them with downstream tasks is a crucial challenge.

C. Transfer Learning

As one of the most important types of transfer learning, domain adaptation leverages knowledge from a label-rich source domain to train a classifier for a label-scarce target domain. Generally, the source and target domains in DA share the same feature space $\mathcal{X}$ and label space $\mathcal{Y}$, but they have different joint probability distributions on the feature-label space $\mathcal{X} \times \mathcal{Y}$. More specially, unsupervised DA, further assumes that only labeled source samples and unlabeled target samples are available.

Existing domain adaptation methods can be divided into two categories: traditional methods and deep learning-based methods [38]. Two major classes of traditional domain adaptation methods are sample-based and feature-based. The sample-based methods reduce the distribution discrepancy between the domains by weighting the samples in the source domain. Such methods include the kernel mean matching (KMM) [39] and TrAdaBoost [40]. TrAdaBoost uses a small amount of labeled data from the target domain and labeled data from the source domain to design a weight strategy according to whether the sample is conducive to the classification. The feature-based methods improve the similarity of the features by learning new feature representations. Pan et al. proposed the transfer component analysis (TCA) [6] method based on the maximum mean discrepancy (MMD) distance of the marginal distribution. TCA reduces the distribution discrepancy between domains by mapping the features to a new latent space. The methods have abundant research, including BDA [7] and MEDA [8].

Deep learning-based methods include discrepancy methods, adversarial methods and feature reconstruction methods. The discrepancy-based methods mainly use the distribution discrepancy measures such as MMD to map the source domain data and the target domain data into a shared feature space [41]. Inspired by generative adversarial networks (GANs), some adversarial-based methods are introduced to find transferable feature representations that are applicable to both source and target domains. Yu et al. [42] proposed a dynamic adversarial adaptation network to quantitatively evaluate the relative importance of domain marginal distribution and conditional distribution through dynamic adversarial distribution alignment. For the methods based on feature reconstruction, they primarily aim to reconstruct the features from both the source and target domains using techniques such as autoencoders or other neural network models. Yang et al. [43] proposed a dual-autoencoder to learn the features of the source domain and the target domain respectively, which can learn the category discrimination information to prevent performance degradation.

Although there has been lots of research on unsupervised domain adaptation in recent years, most of these methods require unlabeled target domain data to learn the invariant representations. However, in real applications, target domain data is often evolving or even unknown to us. Therefore, building representation learning models that can be applied in different target domains only based on source domain data is of great significance to alleviate the problem caused by the inaccessible target data.

To alleviate the aforementioned problem, multiple types of approaches have been proposed, including zero-shot domain adaptation (ZSDA) [44] and DG [12], [13], [45], [46], [47], [48]. ZSDA methods are designed for the case where both data samples and labels in the target domain are unavailable but the dual-domain pair data from source and target domains are available. Therefore, the cost of ZSDA is high due to the challenges associated with collecting the dual-domain pair. In simple terms, DG can be seen as an extension of domain adaptation. It emphasizes learning a model with strong generalization ability for any unseen target domain by leveraging information from multiple source domains. In recent years, many domain generalization methods have been proposed, including MTAE [49] and MMD-AAE [12]. MTAE uses a shared encoder to reconstruct the samples in each domain and then restores them, thereby learning the common features of each sample and reducing the discrepancy of data distributions. MMD-AAE takes data from multiple source domains as input, and addresses the distribution discrepancy between each pair of domains by optimizing the MMD distance. This process enhances the robustness of the learned features by incorporating a discriminator.

Generally, when constructing feature representation for the prediction model on the target domain, DA methods rely on the utilization of sample data from either the target domain or the dual-domain pair data, while DG methods require the collection of data from multiple source domains. Our proposed causal representation method aims to learn invariant representations that can be used to construct cross-domain classification models, even when both samples and label information of the target domain are unknown. This method can address the more challenging scenario of cross-domain classification, where only a single labeled source domain is available.

III. Proposed Method

In this section, we first show the overall framework of GeAE, and then introduce the proposed method in detail.

A. *Overview of the Method*

Given a labeled source domain $\mathcal{D}_s = [\boldsymbol{X}_s, \boldsymbol{Y}_s]$, where $\boldsymbol{X}_s \in \mathbb{R}^{n\times d}$ and $\boldsymbol{Y}_s \in \mathbb{R}^{n\times 1}$ represent the feature variables and labels of samples in the source domain respectively.[1] Our proposed method aims to utilize the knowledge obtained only from a single source domain and learn a robust low-dimensional feature representation model for predicting the label of any target domain data $\mathcal{D}_t = \boldsymbol{X}_t$. It is worth noting that our approach is proposed based on the following assumptions.

Assumption 1: (Causal Graph [32], [50]) For Bayesian Network $< U, \mathbb{G}, P >$, where U denotes the set of variables, $\mathbb{G}$ is a directed acyclic graph (DAG) on U and P denotes the probability distribution of U, then Bayesian Network $< U, \mathbb{G}, P >$ can be used to express the causal relationships between the variables in U. In DAG $\mathbb{G}$, for a pair of directly connected parent-child variables, the parent variable is the direct cause of the child variable, and the child variable is the direct outcome of the parent variable. We assume that the causality between variables can be expressed by $\mathbb{G}$ as a causal graph.

Assumption 2: (Faithfulness [50], [51]) Given a Bayesian network $< U, \mathbb{G}, P >$, DAG $\mathbb{G}$ is said to be faithful to the probability distribution P if and only if each conditional independence relationship in P is dictated by both G and the Markov condition.

Assumption 3: (Causal Sufficiency [27], [50]) If two or more variables in the set U share a common cause, that cause is likewise a member of U. This assumption ensures that the observational data does not contain unobservable latent variables.

The proposed method, GeAE, is built upon a deep autoencoder model that includes a single input layer, multiple hidden layers, and a singular output layer. The framework of GeAE is shown in Fig. 2. Under Assumption 1, the causality in the data can be described by a directed acyclic graph $\mathbb{G}$. Upon obtaining the p-dimensional $(p \le d)$ feature representation, the label variable $\boldsymbol{Y}_s$ is introduced as an additional node to form a new hidden layer. Then, the adjacency matrix $\boldsymbol{A} = \{a_{ij}\}_{(p+1)\times(p+1)}$ associated with $\mathbb{G}$ is incorporated in the autoencoder frame to realize the message passing operation following the causal mechanism. The causal relationships between the variables of the set U in the low-dimensional representation $H^{(l)}$ can be illustrated by the adjacency matrix A. The variables $V_i \in U$ and $V_j \in U$ are considered to be a pair of directly connected parent-child variables if a_{ij} exceeds a specified threshold. Thus, GeAE generates effective low-dimensional causal representations, which can help in making stable predictions for unknown and unlabeled target domains. The objective function of GeAE is defined as follows:

$$\mathcal{L} = \mathcal{L}_{Re} + \lambda_c\mathcal{L}_C + \lambda_y\mathcal{L}_Y + \lambda_r\mathcal{L}_R, \tag{1}$$

where λ_c, λ_y and λ_r are the balancing parameters. The first term $\mathcal{L}_{Re} = \frac{1}{2n}||\boldsymbol{X}_s - \hat{\boldsymbol{X}}_s||^2$ denotes the reconstruction loss, where $\boldsymbol{X}_s$ and $\hat{\boldsymbol{X}}_s$ represent the input and output of the autoencoder, respectively. This term serves as a fundamental component of the autoencoder. The causal structure learning loss, denoted as $\mathcal{L}_C$, is used to uncover the invariant relationships between features and labels. To facilitate this task, a module similar to GAE is integrated into the autoencoder. GeAE improves the performance of causal representations in further classification tasks by incorporating a cross-entropy loss $\mathcal{L}_Y$ into the objective function. The regularization term $\mathcal{L}_R$ is used to prevent the model from overfitting.

[1] In this paper, we use bold capital letters to represent matrices and bold lowercase letters to denote vectors.

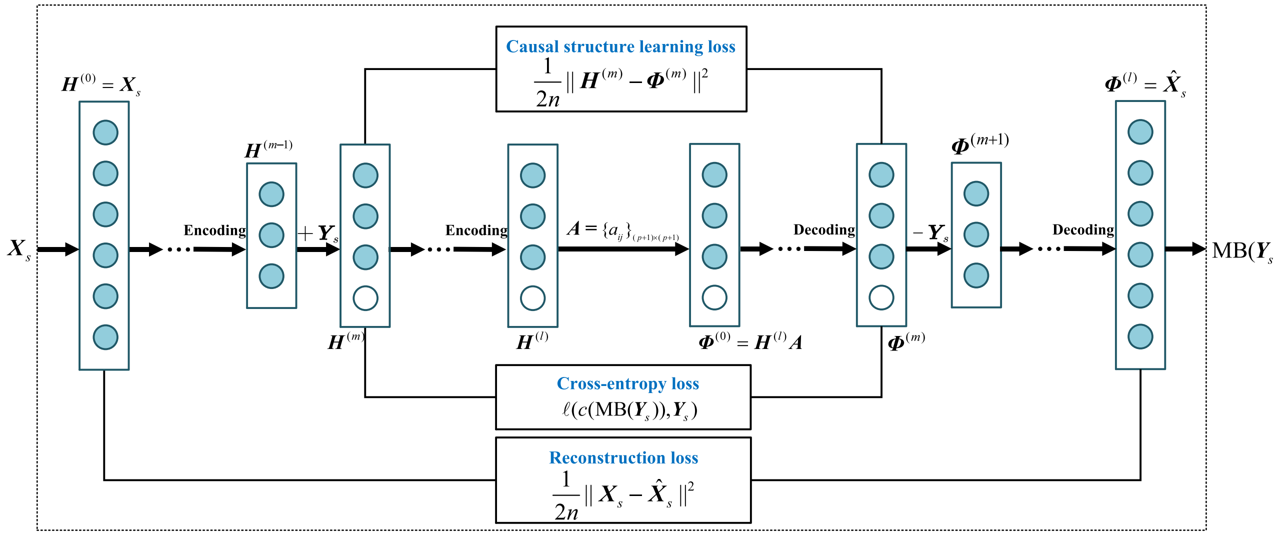


Fig. 2. The framework of GeAE. The input of GeAE is the unlabeled source domain data $\boldsymbol{X}_s$ and the output is $\mathrm{MB}(\boldsymbol{Y}_s)$ representing the low-dimensional causal representation corresponding to the label variable. The adjacency matrix $\boldsymbol{A}$ associated with $\mathbb{G}$ is incorporated in the autoencoder frame to realize the message passing operation following the causal mechanism.

In the following, we will provide comprehensive descriptions of GeAE.

B. Causal Structure Learning Loss $\mathcal{L}_C$

Generally, the goal of causal structure learning is to obtain causal relationships between variables based on the observational data. More recently, multiple gradient-based causal structure learning methods [34], [35], [36] have been proposed, which transform the combinatorial optimization problem of the DAG $\mathbb{G}$ into a continuous one about its adjacent matrix $\boldsymbol{A}$. In addition, to ensure the acyclicity of the causal graph corresponding to the adjacency matrix, Zheng et al. proposed a smooth constraint [34] on $\boldsymbol{A}$:

$$\operatorname{tr}\left(e^{\boldsymbol{A}\odot\boldsymbol{A}}\right)-(p+1)=0, \tag{2}$$

where $\operatorname{tr}(\cdot)$ denotes the trace of a matrix and $e^{\boldsymbol{M}}$ denotes the matrix exponential of $\boldsymbol{M}$; $\odot$ is the Hadamard product.

Lemma 1: Given a graph $\mathbb{G}=\{U,E\}$ and its corresponding adjacency matrix $\boldsymbol{A}\in\mathbb{R}^{(p+1)\times(p+1)}$, the element $a_{ij}^{(k)}$ in the matrix $\boldsymbol{A}^k$ (the k_{th} power of $\boldsymbol{A}$) represents the number of paths of length k from node V_i to node V_j $i,j=1,\ldots,p+1$.

Theorem 1: The adjacent matrix $\boldsymbol{A}$ of a DAG $\mathbb{G}$ is acyclic if and only if (2) holds.

Proof: Let $\boldsymbol{S}=\boldsymbol{A}\odot\boldsymbol{A}$. It is easy to see $\boldsymbol{S}$ is nonnegative, i.e.,

$$\boldsymbol{S}\in\mathbb{R}_+^{(p+1)\times(p+1)}.$$

First, we prove the sufficiency. If (2) holds, we have

$$\begin{aligned}\operatorname{tr}\left(e^{\boldsymbol{S}}\right)-(p+1)&=\operatorname{tr}(\boldsymbol{I})-(p+1)+\operatorname{tr}(\boldsymbol{S})+\frac{1}{2!}\operatorname{tr}\left(\boldsymbol{S}^2\right)+\cdots\\&=\operatorname{tr}(\boldsymbol{S})+\frac{1}{2!}\operatorname{tr}\left(\boldsymbol{S}^2\right)+\cdots=0.\end{aligned} \tag{3}$$

As $\boldsymbol{S}\in\mathbb{R}_+^{(p+1)\times(p+1)}$, we can get $\operatorname{tr}(\boldsymbol{S}^k)=0$ for all $k\in\mathbb{N}^*$. It means $s_{ii}^{(k)}=0\ (i=1,2,\ldots,p+1)$. By Lemma 1, there are no k-cycles in graph $\mathbb{G}$, proving that $\mathbb{G}$ is acyclic.

Next, we prove the necessity. If $\mathbb{G}$ is acyclic, by Lemma 1 we can know, $s_{ii}^{(k)}=0\ (i=1,2,\ldots,p+1)$, which implies $\operatorname{tr}(\boldsymbol{S}^k)=0$. Thus, we have

$$\begin{aligned}0&=\operatorname{tr}(\boldsymbol{S})+\frac{1}{2!}\operatorname{tr}\left(\boldsymbol{S}^2\right)+\cdots\\&=\operatorname{tr}(\boldsymbol{S})+\frac{1}{2!}\operatorname{tr}(\boldsymbol{S}^2)+\cdots+\operatorname{tr}(\boldsymbol{I})-(p+1)\\&=\operatorname{tr}(e^{\boldsymbol{S}})-(p+1).\end{aligned} \tag{4}$$

The necessity is proven. □

In the following, we introduce how to design the causal structure learning loss $\mathcal{L}_C$ in GeAE. As an autoencoder model, GeAE consists of two modules: encoder and decoder. Given the input data $\boldsymbol{X}_s\in\mathbb{R}^{n\times d}$, the encoder can learn low-dimensional representations with dimension p $(p\leq d)$. Then the decoder can output the estimate of $\boldsymbol{X}_s$, i.e., $\hat{\boldsymbol{X}}_s$, based on the low-dimensional representations. The encoder-decoder process can be summarized as follows:

Encoder:

$$\boldsymbol{H}^{(0)}=\boldsymbol{X}_s; \tag{5}$$

$$\boldsymbol{H}^{(i)}=\sigma(\boldsymbol{H}^{(i-1)}\boldsymbol{w}_1^{(i)}+\boldsymbol{b}_1^{(i)}),i=1,\ldots,m-1; \tag{6}$$

$$\boldsymbol{H}^{(m)}=\left[\boldsymbol{H}^{(m-1)},\boldsymbol{Y}_s\right]; \tag{7}$$

$$\boldsymbol{H}^{(i)}=\sigma\left(\boldsymbol{H}^{(i-1)}\boldsymbol{w}_1^{(i)}+\boldsymbol{b}_1^{(i)}\right),i=m+1,\ldots,l. \tag{8}$$

Decoder:

$$\boldsymbol{\Phi}^{(0)}=\boldsymbol{H}^{(l)}\boldsymbol{A}; \tag{9}$$

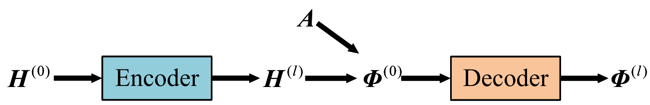


Fig. 3. Causal structure learning module in GeAE.

$$\boldsymbol{\Phi}^{(i)} = \sigma\left(\boldsymbol{\Phi}^{(i-1)}\boldsymbol{w}_2^{(i)} + \boldsymbol{b}_2^{(i)}\right), i = 1, \ldots, m; \tag{10}$$

$$\boldsymbol{\Phi}^{(m+1)} = \boldsymbol{\Phi}^{(m)}\left[:, 1 : (p-1)\right]; \tag{11}$$

$$\boldsymbol{\Phi}^{(i)} = \sigma\left(\boldsymbol{\Phi}^{(i-1)}\boldsymbol{w}_2^{(i)} + \boldsymbol{b}_2^{(i)}\right), i = m+2, \ldots, l. \tag{12}$$

Here, l denotes the number of hidden layers, and σ denotes nonlinear activation functions. $M[:, 1 : (p-1)]$ denotes the matrix composed of the first $p-1$ columns. Let $\boldsymbol{\Phi}^{(l)} = \hat{\boldsymbol{X}}_s$. $\boldsymbol{w}_1^{(i)}$ and $\boldsymbol{b}_1^{(i)}$ denote the weight matrix and bias vector on the i^{th} encoding layer, respectively. Similarly, $\boldsymbol{w}_2^{(i)}$ and $\boldsymbol{b}_2^{(i)}$ denote the weight matrix and bias vector on the i^{th} decoding layer, respectively. $\boldsymbol{H}^{(m-1)} \in \mathbb{R}^{n\times p}$ in the encoding process can be seen as the low-dimensional representations of $\boldsymbol{X}_s$.

To identify the causal relationship between the feature variables and target labels, the hidden layer must include the label information $\boldsymbol{Y}_s$ for the source domain data. Thus, starting with low-dimensional feature representation $\boldsymbol{H}^{(m-1)}$, GeAE incorporates $\boldsymbol{Y}_s$ as an additional node to construct a new hidden layer $\boldsymbol{H}^{(m)}$ (7). Subsequently, $\boldsymbol{Y}_s$ is removed after the m^{th} decoding layer (11).

The computational process from $\boldsymbol{H}^{(m)}$ to $\boldsymbol{\Phi}^{(m)}$ can be written as follows:

$$\begin{aligned} \boldsymbol{H}^{(l)} &= g_1\left(\boldsymbol{H}^{(m)}\right), \\ \boldsymbol{\Phi}^{(0)} &= \boldsymbol{H}^{(l)}\boldsymbol{A}, \\ \boldsymbol{\Phi}^{(m)} &= g_2\left(\boldsymbol{\Phi}^{(0)}\right). \end{aligned} \tag{13}$$

Here, g_1 and g_2, which can be regarded as two nonlinear functions, are used to describe the encoding and decoding process respectively. Based on (13), we obtain the following relationship between $\boldsymbol{\Phi}^{(m)}$ and $\boldsymbol{H}^{(m)}$:

$$\begin{aligned} \boldsymbol{\Phi}^{(m)} &= g_2(g_1(\boldsymbol{H}^{(m)})\boldsymbol{A}) \\ &= g(\boldsymbol{H}^{(m)}, \boldsymbol{A}), \end{aligned} \tag{14}$$

where g can be viewed as a nonlinear function.

Inspired by [36], the causal structure learning loss $\mathcal{L}_C$ can be defined as

$$\mathcal{L}_C = \frac{1}{2n}||\boldsymbol{H}^{(m)} - g(\boldsymbol{H}^{(m)}, \boldsymbol{A})||^2, \tag{15}$$

where the adjacent matrix $\boldsymbol{A}$ is required to satisfy the constraint (2).

It is noted here that the causal structure learning module can be seen as a graph autoencoder shown in Fig. 3. Based on (14), we can rewrite (15) as

$$\mathcal{L}_C = \frac{1}{2n}||\boldsymbol{H}^{(m)} - \boldsymbol{\Phi}^{(m)}||^2. \tag{16}$$

Equation (16) can be considered as the reconstruction loss of a GAE. From (13), we can see that g_1 and g_2 can be viewed as the encoder and decoder of GAE respectively. The latent representation layer $\boldsymbol{H}^{(l)}$ can be extracted by the encoder g_1 which can be used to represent the other latent layer $\boldsymbol{\Phi}^{(0)}$ by

$$\boldsymbol{\Phi}^{(0)} = \boldsymbol{H}^{(l)}\boldsymbol{A}. \tag{17}$$

The linear message passing operation shown in (17) between the latent layers $\boldsymbol{H}^{(l)}$ and $\boldsymbol{\Phi}^{(0)}$ is similar to the graph convolutional layer [36], [52], where the activation function is linear and the weight matrix is the identity matrix.

Once $\boldsymbol{A}$ is determined, we can identify the unique Markov blanket of the target label MB$(\boldsymbol{Y}_s)$, which includes its parents, children and spouses under the faithfulness assumption and the causal sufficiency assumption [27], [53]. It is noted here that even after passing through multiple fully connected layers from $\boldsymbol{H}^{(m)}$ to $\boldsymbol{\Phi}^{(m)}$, the last dimension of the representation in each intermediate layer continues to hold the label information based on the message passing mechanism in $\boldsymbol{A}$ [36]. MB$(\boldsymbol{Y}_s)$ is called the causal representation. Then, the low-dimensional representations in $\boldsymbol{H}^{(m)}$ can be divided into two groups: causal representations MB$(\boldsymbol{Y}_s)$, and the other feature representations which may be task-irrelevant features.

C. *Cross-Entropy Loss* $\mathcal{L}_Y$

The causal relationships between the label variable and its corresponding MB features remain robust across various contexts. This robustness suggests that these relationships can be considered invariant across different domains, thereby enabling the development of a predictive model applicable to unknown target domains. The following objective function is constructed to improve the quality of the predictions using the causal representations:

$$\mathcal{L}_Y = \ell\left(c\left(\text{MB}\left(\boldsymbol{Y}_s\right)\right), \boldsymbol{Y}_s\right), \tag{18}$$

where $\ell(\cdot)$ denotes the cross-entropy loss and $c(\cdot)$ denotes a classifier. It is noteworthy that the causal structure learning loss and the cross-entropy loss jointly contribute to the optimization of adjacency matrix $\boldsymbol{A}$ and the effectiveness of the representation features for the classification task.

D. *Regularization Loss* $\mathcal{L}_R$

In addition, to prevent overfitting and performance degradation of the trained model, we design a regularization term $\mathcal{L}_R$ on the weight parameters to the objective function $\mathcal{L}$:

$$\mathcal{L}_R = \sum_{\substack{i=1 \\ i\neq m}}^{l}\left(||\boldsymbol{w}_1^{(i)}||^2\right) + \sum_{\substack{i=2 \\ i\neq m+1}}^{l+1}\left(||\boldsymbol{w}_2^{(i)}||^2\right). \tag{19}$$

Remark: Since each bias parameter controls only a single variable, not regularizing it does not lead to significant variance. Furthermore, regularizing bias parameters can result in noticeable underfitting. Therefore, in neural networks, regularization is typically applied only to weight parameters, not bias parameters [54]. Also, in this paper, the regularization term L_R does not impose constraints on bias terms.

E. Optimization

According to (16), (18) and (19), the objective function of GeAE can be defined as

$$\mathcal{L} = \frac{1}{2n}||\boldsymbol{X}_s - \hat{\boldsymbol{X}}_s||^2 + \frac{\lambda_c}{2n}||\boldsymbol{H}^{(m)} - \boldsymbol{\Phi}^{(m)}||^2 + \lambda_y \cdot \ell(c(\mathrm{MB}(\boldsymbol{Y}_s)), \boldsymbol{Y}_s) + \lambda_r \left[\sum_{\substack{i=1 \\ i \neq m}}^{l} (||\boldsymbol{w}_1^{(i)}||^2) + \sum_{\substack{i=2 \\ i \neq m+1}}^{l+1} (||\boldsymbol{w}_2^{(i)}||^2) \right]. \tag{20}$$

The above objective function should be minimized with constraint in (2). We will use the augmented Lagrangian method [36], [55] to solve the equality-constrained program. The augmented Lagrangian is given by

$$\mathcal{L}(\boldsymbol{A}, \alpha, \rho) = \frac{1}{2n}||\boldsymbol{X}_s - \hat{\boldsymbol{X}}_s||^2 + \frac{\lambda_c}{2n}||\boldsymbol{H}^{(m)} - \boldsymbol{\Phi}^{(m)}||^2 + \lambda_y \cdot \ell(c(\mathrm{MB}(\boldsymbol{Y}_s)), \boldsymbol{Y}_s) + \lambda_r \left[\sum_{\substack{i=1 \\ i \neq m}}^{l} (||\boldsymbol{w}_1^{(i)}||^2) + \sum_{\substack{i=2 \\ i \neq m+1}}^{l+1} (||\boldsymbol{w}_2^{(i)}||^2) \right] + \alpha h(\boldsymbol{A}) + \frac{\rho}{2}|h(\boldsymbol{A})|^2, \tag{21}$$

where α denotes the Lagrange multiplier, $\rho > 0$ denotes the penalty parameter and $h(\boldsymbol{A}) = \mathrm{tr}(e^{\boldsymbol{A} \odot \boldsymbol{A}}) - (p+1)$. Different from the traditional Lagrangian method, the augmented Lagrangian incorporates a quadratic penalty term for the nonlinear constraint $h(\boldsymbol{A})$, which can improve convergence speed and numerical stability [55], [56]. We then have the following update rules for the adjacent matrix $\boldsymbol{A}$, α and ρ:

$$\boldsymbol{A}^{(t+1)} = \arg\min \ \mathcal{L}\left(\boldsymbol{A}^{(t)}, \alpha^{(t)}, \rho^{(t)}\right), \tag{22}$$

$$\alpha^{(t+1)} = \alpha^{(t)} + \rho^{(t)} h\left(\boldsymbol{A}^{(t+1)}\right), \tag{23}$$

$$\rho^{(t+1)} = \begin{cases} \beta \rho^{(t)}, & \text{if } |h(\boldsymbol{A}^{(t+1)})| \leq \theta |h(\boldsymbol{A}^{(t)})|, \\ \rho^{(t)}, & \text{otherwise}, \end{cases} \tag{24}$$

where t denotes the t^{th} iteration; $\beta > 1$ and $\theta < 1$ are two tuning hyperparameters. The gradient descent method is applied to solve the problem (22). After obtaining the optimal $\boldsymbol{A}$, we can get the causal representation by finding $\mathrm{MB}(\boldsymbol{Y}_s)$ based on $\boldsymbol{H}^m$ and $\boldsymbol{A}$. The proposed GeAE algorithm is shown in Algorithm 1.

F. Computational Complexity

The computational complexity of GeAE mainly comes from the deep autoencoder model. Given the number of hidden layers l and the number of hidden layer nodes q, the computational complexity of GeAE is $O(nd^2q^{2l-3}T)$, where n and d denote the number of samples and feature dimensions of the input $\boldsymbol{X}_s$, and T denotes the number of iterations.

Algorithm 1: GeAE.

Input: Source domain data $[\boldsymbol{X}_s, \boldsymbol{Y}_s]$; parameters λ_c, λ_y, λ_r, l, m, β, θ; the maximum number of iterations T;
Output: The causal representation $\mathrm{MB}(\boldsymbol{Y}_s)$;
1: Initialization: Initial $w_1^{(i)}, b_1^{(i)}, i = 1, \ldots, l, i \neq m$ and $w_2^{(i)}, b_2^{(i)}, i = 2, \ldots, l+1, i \neq m+1$ randomly. Let $\boldsymbol{A} = \mathrm{I}, \rho = 1, \alpha = 0, t = 0$.
2: **while** $t \leq T$ **do**
3: Update the adjacent matrix $\boldsymbol{A}^{(t+1)}$ by (22);
4: Update $\alpha^{(t+1)}$ and $\rho^{(t+1)}$ by (23) and (24), respectively;
5: $t \leftarrow t+1$;
6: **end while**
7: Extract the causal representation $\mathrm{MB}(\boldsymbol{Y}_s)$ by $\boldsymbol{A}$ and $\boldsymbol{H}^{(m)}$.
8: **return** $\mathrm{MB}(\boldsymbol{Y}_s)$

IV. Experiment

In this section, we will compare the proposed method GeAE[2] with seven feature representation methods, including five causality-based methods (CAE[3] , GAE[4] [36], GSMB [57], IAMB [58] and BAMB[5]) and two classical feature selection methods (FCBF [23] and mRMR [24]). All experiments are conducted on a computer with Windows 10, Intel(R) i5-10400, 2.90 GHz CPU, and 16 GB of memory.

For GeAE, set the number of hidden layers to be $l = 5$ and each hidden layer to have 50 neurons. Let

$$T = 10, m = 3, \beta = 10, \theta = 0.25.$$

For other comparison methods, all the parameters are set as default. We initially acquire feature representations from the source domains using GeAE and other comparison methods. Subsequently, a support vector machine (SVM) classifier is trained by utilizing the obtained representations along with the labels available in the source domain. Finally, we proceed with the classification task across various target domains. The classification accuracy on the target domain is adopted as the evaluation metric. For GeAE and CAE, the algorithms are repeated 5 times on each task and the average accuracy is reported. It is important to note that we have not directly compared GeAE with domain adaptation methods due to the requirement of the target domain samples for such methods. GeAE, on the other hand, is specifically designed to address a more challenging task where the target data is entirely unknown.

[2] The code is available at https://github.com/MingJiang123/GeAE.

[3] The code is available at https://github.com/kuiy/Causal-Autoencoder-CAE-.

[4] The code is available at https://github.com/zhushy/trustworthyAI-1. Generally, "GAE" refers to a graph autoencoder. In [36], though, the authors use it for their proposed causal structure learning method based on a graph autoencoder. To maintain consistency, we adopt the notation "GAE" in this section to denote their method.

[5] The codes are available at https://github.com/kuiy/pyCausalFS.

A. Synthetic Data

First, we conduct experiments on two kinds of synthetic datasets to demonstrate the superiority of GeAE in terms of learning nonlinear causal relationships. Similar to [36], we generate the nonlinear and linear datasets by sampling from a nonlinear causal structure equation and a linear one with a random DAG $\mathbb{G}$ respectively. Assume there are n_s samples in the source domain and n_t samples in the target domain, where the dimension of the data in both domains is d. Here we consider a scenario where the samples in each domain are partitioned into two distinct subgroups, which are denoted by

$$\boldsymbol{X}_{s_1}, \boldsymbol{X}_{s_2} \in \mathbb{R}^{(n_s/2)\times d}$$

for the source and

$$\boldsymbol{X}_{t_1}, \boldsymbol{X}_{t_2} \in \mathbb{R}^{(n_t/2)\times d}$$

for the target. For each domain, the samples in the two subgroups are initialized by different multivariate Gaussian distributions. If the graph nodes are sorted in the topological order, the equations for generating source and target datasets can be designed by (25) and (26) respectively:

$$\boldsymbol{X}_{s_i}^{(j)} \leftarrow \boldsymbol{X}_{s_i}^{(j)} + f\left(\mathrm{Pa}\left(\boldsymbol{X}_{s_i}^{(j)}\right)\right) + \epsilon_1, j = 1, \ldots, d, i = 1, 2; \tag{25}$$

$$\boldsymbol{X}_{t_i}^{(j)} \leftarrow \boldsymbol{X}_{t_i}^{(j)} + f\left(\mathrm{Pa}\left(\boldsymbol{X}_{t_i}^{(j)}\right)\right) + \epsilon_2, j = 1, \ldots, d, i = 1, 2, \tag{26}$$

where $\mathrm{Pa}(x)$ denotes the parents of variable x in $\mathbb{G}$. The source domain features and target domain features are constructed by iterating on each feature with the above equations. Assume that ϵ_1 and ϵ_2 follow different Gaussian distributions with

$$\epsilon_1 \sim N(0, 1)$$

and

$$\epsilon_2 \sim N(0.5, 1)$$

respectively. They are utilized for simulating distinct environments in the source and target domains. In this experiment, for the nonlinear dataset, we set

$$f(x) = \cos(x) + 1,$$

while for the linear dataset, we use

$$f(x) = 0.2 \times x + 1.$$

We conduct experiments on four different feature dimensions with $d \in \{50, 100, 150, 200\}$. Let the sample size be $n_s = n_t = n = 1000$. The values of λ_c, λ_y and λ_r are all set to 1. The classification results on the nonlinear and linear datasets using the causal features obtained from GeAE and the comparison methods are reported in Tables I and II. Underlined values in the tables indicate the highest accuracy for each task and the average. This format is used consistently across all tables showing accuracy in the following. Compared with the other methods, GeAE outperforms other methods on the nonlinear and linear datasets in terms of average accuracy, which indicates the significance of causal representations for cross-domain classification. Overall, GeAE, CAE, and GAE perform better than other methods across all feature dimensions. However, for nonlinear datasets, GeAE and GAE demonstrate a clearer advantage over CAE, which can be attributed to the graph autoencoder structure's effectiveness in handling such data.

TABLE I
ACCURACY (%) ON THE NONLINEAR SYNTHETIC DATASETS

Method	d				Average
	50	100	150	200	
FCBF	50.30	55.90	58.80	55.40	55.10
mRMR	73.20	70.80	78.10	75.30	74.35
GSMB	63.70	71.00	74.10	78.50	71.83
IAMB	61.50	71.20	75.30	78.00	71.50
BAMB	65.20	71.50	79.20	78.70	73.65
CAE	61.30	74.10	79.10	77.70	73.05
GAE	<u>78.40</u>	79.90	81.30	80.50	80.03
GeAE	78.30	<u>81.90</u>	<u>81.70</u>	<u>81.30</u>	<u>80.80</u>

TABLE II
ACCURACY (%) ON THE LINEAR SYNTHETIC DATASETS

Method	d				Average
	50	100	150	200	
FCBF	57.70	50.90	58.70	49.70	54.25
mRMR	74.70	77.40	76.80	72.00	75.23
GSMB	62.70	71.90	74.30	73.10	70.50
IAMB	64.10	71.80	73.50	71.40	70.20
BAMB	64.10	71.90	73.20	72.10	70.33
CAE	76.80	74.90	76.90	78.00	76.65
GAE	<u>79.10</u>	80.90	82.70	<u>79.40</u>	80.53
GeAE	77.80	<u>84.10</u>	<u>82.80</u>	79.00	<u>80.93</u>

To further illustrate the efficiency of GeAE, we compare the running times of CAE and GeAE on the nonlinear datasets. The results show that the running time of GeAE with different feature dimensions is much smaller than that of CAE (see Fig. A1 in Appendix, available online). It indicates that GeAE is more efficient. In addition, we use t-distributed Stochastic Neighbor Embedding (t-SNE) to visualize the invariant representations on the nonlinear datasets obtained by GeAE and CAE respectively. The visualization results on the datasets with four feature dimensions are shown in Figs. A2– A5 in Appendix respectively, available online.

B. Case Study

To illustrate the effectiveness of GeAE in leveraging low-dimensional representations, particularly in extracting features with high causality while eliminating spurious features, we design an experiment on synthetic datasets with varying ratios of causal and spurious features.

In this analysis, we focus on datasets that encompass two distinct classes within each domain. Each class comprises 500 samples, with each sample characterized by a consistent dimensionality of $d = 200$. The causal features for the source and target domain data are generated based on the following causal structure equations:

$$\boldsymbol{X}_{s_i}^{(j)} \leftarrow \boldsymbol{X}_{s_i}^{(j)} + f\left(\mathrm{Pa}\left(\boldsymbol{X}_{s_i}^{(j)}\right)\right) + \epsilon, j = 1, \ldots, n_c, i = 0, 1; \tag{27}$$

TABLE III
RESULTS ON DATASETS WITH DIFFERENT VALUES OF RC

RC (%)	0	50	75	100
Accuracy	16.6	78.00	90.30	98.60

$$\boldsymbol{X}_{t_i}^{(j)} \leftarrow \boldsymbol{X}_{t_i}^{(j)} + f\left(\mathrm{Pa}\left(\boldsymbol{X}_{t_i}^{(j)}\right)\right) + \epsilon, j = 1, \ldots, n_c, i = 0, 1, \tag{28}$$

where $f(x) = \cos(x) + 1$, n_c is the dimension of causal features in the dataset, t_i (or s_i) where $i = 0, 1$ represent the two classes of samples in the target (or source) domain, and ϵ represents Gaussian noise. During data generation, the ratios of causal features (RC) are predefined, such that $n_c = 200 \times \mathrm{RC}$. By varying RC across 0% , 50% , 75% , and 100% , we create four datasets, each with a different proportion of causal features. The spurious features are added by sampling from Gaussian distributions. Two independent Gaussian features with different parameters, z_0 and z_1, are then added to the samples in each domain. The new source and target domain data are denoted by

$$\boldsymbol{X}_{s_i} \leftarrow \left[\boldsymbol{X}_{s_i}, \{z_i\}_{500\times(d-n_c)}\right],\ i = 0, 1, \tag{29}$$

and

$$\boldsymbol{X}_{t_i} \leftarrow \left[\boldsymbol{X}_{t_i}, \{z_{1-i}\}_{500\times(d-n_c)}\right],\ i = 0, 1. \tag{30}$$

We assume that z_0 and z_1 follow different Gaussian distributions, specified as

$$z_0 \sim \mathcal{N}(1, \sigma_1^2),\ z_1 \sim \mathcal{N}\left(1, \sigma_2^2\right),$$

where σ_1 is randomly chosen from $[0, 1]$, while σ_2 is selected from $[0.5, 1.5]$. The results on the four generated datasets are presented in Table III. From the table, we observe that as the proportion of causal features increases, the performance of GeAE significantly improves. This can be attributed to its ability to effectively extract features with high causality while eliminating spurious features.

C. Real-World Data

In this experiment, we will test with four commonly used cross-domain classification datasets, including Amazon Review, Reuters-21578, Office-Caltech10 and PaperCitation. Amazon Review is a standard benchmark for cross-domain sentiment analysis. It contains product reviews from four domains: DVD (D), Book (B), Electronic (E) and Kitchen (K). Each domain consists of about 1000 positive samples and 1000 negative samples. For fair comparison, we adopt the preprocessed version of Amazon Review [22] and generate 12 cross-domain tasks. Reuters-21578 is a text dataset consisting of many subcategories. We construct 6 cross-domain tasks by selecting the three largest top categories: Orgs (Or), People (Pe) and Places (Pl). The preprocessed version of Reuters-21578 [59] with 4771 features is adopted in this experiment. Office-caltech10 contains 2533 images from 10 categories in four domains: Amazon (A), Webcam (W), Dslr (D) and Caltech256 (C) which is widely used in domain adaptation [60], [61] and domain generalization [62]. The SURF version with 800 dimensional features is used for the experiment and 12 cross-domain tasks are constructed. The PaperCitation dataset consists of two citation networks from different sources: ACM and DBLP. The features from the first 200 dimensions are used to create two cross-domain node classification tasks. The detailed information of these datasets is shown in Table A1 in the appendix, available online. In the implementation of GeAE, the values of λ_c, λ_y and λ_r are all set to 1.

For these four datasets, the accuracy of the classification results for the target domain data based on representational features obtained by GeAE and the other seven comparison methods are reported in Tables IV, V, VI, and VII. The tables show that for the Amazon Review dataset, GeAE and CAE significantly outperform other methods, with GeAE demonstrating a slight advantage over CAE in terms of the average performance. For the Reuters-21578 dataset, GeAE achieves the highest average classification accuracy among all eight methods. This shows the superiority of GeAE for causal structure learning with high-dimensional data. We note that for this dataset, only the first 400 feature dimensions are used for GAE due to its high complexity. For the Office-Caltech10 dataset, GeAE and CAE are the most effective methods, with CAE showing a marginally higher average accuracy than GeAE. For the PaperCitation dataset, GeAE achieves better node classification accuracy in both tasks compared to the other methods, demonstrating its effectiveness on graph datasets. Overall, in terms of average accuracy, GeAE ranks first in three datasets and second in one. From the experimental results above, we can summarize the following conclusions:

- Compared to traditional feature representation methods, GeAE outperforms FCBF and mRMR. This demonstrates the necessity of causal representation learning in cross-domain tasks and showcases its effectiveness in identifying causal relationships.
- Compared to causal representation methods, GeAE is superior to GSMB, IAMB, BAMB and CAE. Since GeAE is based on the graph autoencoder framework, it can more effectively capture causal representations in high-dimensional nonlinear datasets.
- GeAE builds on the causal learning approach GAE, which is not designed for classification tasks. The superior performance of GeAE compared with GAE suggests that adding cross-entropy loss improves the effectiveness of causal representations for cross-domain classification.

In order to show the results of GeAE on the real-world datasets more clearly, we use t-SNE to visualize the invariant representation of different target domains obtained by GeAE based on a given source domain (see Figs. A6– A9 in Appendix, available online). From these scatter plots, it can be observed that the representations obtained by GeAE from a fixed source domain can exhibit a high level of data separation across multiple different target domains. This indicates that GeAE has good generalization performance. Furthermore, the running time of GeAE and the comparison methods on each domain of different datasets is recorded (see Tables A2– A5 in Appendix, available online). As observed, the running time of GeAE is slightly higher than traditional methods. However, compared to CAE and GAE, GeAE significantly reduces running time

TABLE IV
ACCURACY (%) OF THE 12 CROSS-DOMAIN TASKS ON THE AMAZON REVIEW DATASET

Methods	B→D	B→E	B→K	D→B	D→E	D→K	E→B	E→D	E→K	K→B	K→D	K→E	Average
FCBF	52.23	48.15	51.38	56.10	53.80	55.23	56.50	58.98	63.33	54.65	52.03	55.16	54.80
mRMR	68.93	66.47	66.08	65.20	66.62	67.53	63.60	67.28	76.04	64.55	68.18	73.07	67.80
GSMB	52.93	51.50	51.83	50.00	50.05	50.03	50.00	50.03	50.03	50.00	50.03	50.05	50.54
IAMB	60.58	57.86	60.78	50.00	50.05	50.03	50.00	50.03	50.03	50.00	50.03	50.05	52.45
BAMB	53.73	55.26	53.78	53.75	53.50	53.63	56.25	58.83	62.43	56.35	58.63	62.31	56.54
CAE	76.44	72.52	75.14	75.80	73.17	76.09	72.85	73.59	82.79	74.20	74.43	83.18	75.85
GAE	55.33	52.90	52.68	50.00	50.05	50.03	50.00	50.03	50.03	50.00	50.03	50.05	50.93
GeAE	76.69	76.13	78.94	71.25	75.73	76.94	70.05	73.49	84.39	73.55	73.44	81.63	76.02

TABLE V
ACCURACY (%) OF THE 6 CROSS-DOMAIN TASKS ON THE REUTERS-21578 DATASET

Methods	Or → Pl	Pl → Or	Or → Pe	Pe → Or	Pl → Pe	Pe → Pl	Average
FCBF	65.35	57.64	65.23	58.29	64.98	57.87	58.76
mRMR	50.10	54.24	73.34	72.35	57.70	55.31	60.51
GSMB	63.09	60.14	62.42	61.28	57.94	49.03	58.98
IAMB	60.88	59.40	65.81	62.17	59.24	49.58	59.51
BAMB	60.98	55.31	55.63	48.83	57.10	58.50	56.06
CAE	56.09	57.87	51.41	52.47	57.66	60.26	55.96
GAE	61.94	66.73	73.92	62.17	58.87	57.57	63.53
GeAE	72.21	63.63	74.54	77.43	58.21	57.31	67.22

TABLE VI
ACCURACY (%) OF THE 12 CROSS-DOMAIN TASKS ON THE OFFICE-CALTECH10 DATASET

Methods	C→A	C→W	C→D	A→C	A→W	A→D	W→C	W→A	W→D	D→C	D→A	D→W	Average
FCBF	29.65	22.93	20.34	25.91	24.75	23.57	22.62	19.00	40.76	19.15	16.49	38.64	25.32
mRMR	37.47	30.51	33.76	25.56	26.10	31.21	19.95	18.16	57.96	23.06	23.49	50.51	31.48
GSMB	34.86	23.73	25.48	30.28	28.14	26.11	25.91	27.77	41.40	20.66	17.33	17.29	26.50
IAMB	30.58	20.00	28.66	28.14	27.19	18.47	19.15	23.70	45.86	20.04	21.71	28.14	25.97
BAMB	24.43	19.32	22.29	25.73	23.05	26.75	16.83	17.12	25.48	16.92	10.44	25.08	21.12
CAE	48.85	42.71	42.68	43.46	39.67	36.16	33.04	37.89	81.53	32.98	34.32	78.12	45.94
GAE	20.25	18.31	18.47	30.63	29.30	22.37	18.88	20.56	24.84	13.54	11.38	15.59	20.34
GeAE	51.82	34.10	44.59	43.22	34.03	37.20	33.34	36.20	82.55	32.79	33.55	76.61	45.00

TABLE VII
ACCURACY (%) OF THE 2 CROSS-DOMAIN TASKS ON PAPERCITATION DATASET

Methods	ACM → DLBP	DLBP → ACM	Average
FCBF	18.59	15.55	17.07
mRMR	18.90	17.05	17.98
GSMB	26.43	31.48	29.00
IAMB	26.63	32.12	29.38
BAMB	26.48	31.73	29.10
CAE	16.25	6.30	11.28
GAE	19.92	18.72	19.32
GeAE	37.56	43.88	40.72

on real datasets, highlighting the benefits of integrating causal structure learning with feature reconstruction in a unified graph autoencoder framework.

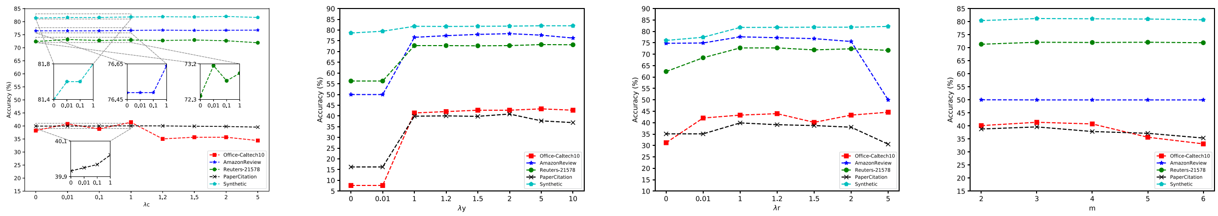
In summary, from the experimental results on synthetic and real-world datasets, GeAE is superior to the state-of-the-art methods, especially when the dataset is of high dimension and nonlinear causality. GeAE enables the effective application of causal representations obtained from a single source domain to construct a predictive model for an unseen target domain.

D. Parameter Sensitivity Analysis

To investigate the sensitivity to the balanced parameters λ_c, λ_y and λ_r, we conduct experiments to perform parameter sensitivity analysis on the non-linear synthetic dataset with $d = 150$, and the four real-world datasets. For Amazon Review, the task B → D is selected; for Reuters-21578, the task Or → Pl is chosen; for Office-Caltech10, the task C → D is opted; and for Papercitation, we select the task ACM → DLBP.

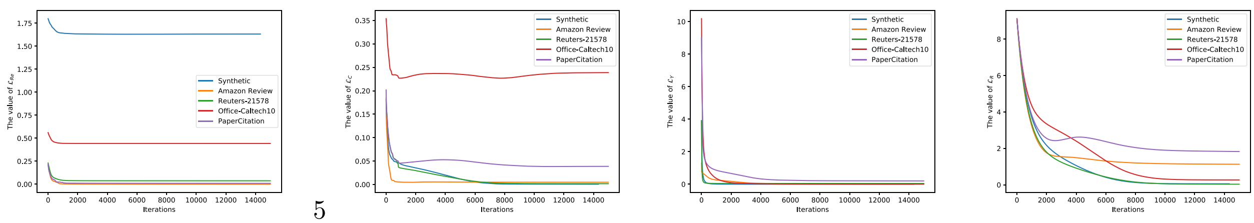
Let $\lambda_y = \lambda_r = 1$, the classification accuracy results with difference values of λ_c are shown in Fig. 4(a). Similarly, setting the values of the other two parameters to be 1, the results with difference values of λ_y and λ_r are displayed in Fig. 4(b) and (c) respectively. As shown in the figures, GeAE is sensitive to the parameters λ_c, λ_y and λ_r. Specifically, for λ_c, the optimal value range is [0.01,1] for Office-Caltech10, and [0.01,1.5] for the other datasets. Regarding λ_y, the optimal range is [1,2] across all the datasets. As for λ_r, the range of [1,2] is optimal for the synthetic dataset, as well as the Amazon Review, Reuters-21578 and PaperCitation datasets, while for Office-Caltech10, the optimal range is [1,1.2]. Based on these findings, the values of λ_c, λ_y and λ_r can be set to 1in the implementation of GeAE.

We also investigate the sensitivity of GeAE with the model structure parameter m, which affects the number of the hidden layers in GeAE. The results of the classification accuracy with different values of $m \in \{2, 3, 4, 5, 6\}$ are shown in Fig. 4(d). We can find that the accuracy improves lightly with the increasing of m. However, the increase of m also results in a decrease



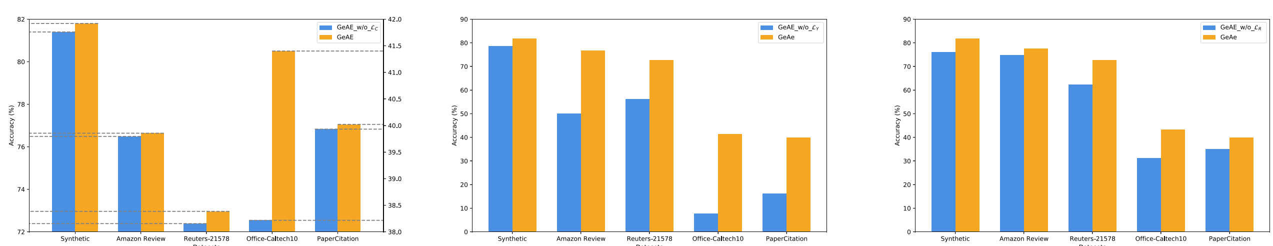
(a) Results with different λ_c. (b) Results with different λ_y. (c) Results with different λ_r. (d) Results with different m.

Fig. 4. Parameters sensitivity study on all the datasets for GeAE.



(a) $\mathcal{L}_{Re}$. (b) $\mathcal{L}_C$. (c) $\mathcal{L}_Y$. (d) $\mathcal{L}_R$.

Fig. 5. The trajectory of each component of the loss function.



(a) Results of GeAE and GeAE_w/o_$\mathcal{L}_C$. (b) Results of GeAE and GeAE_w/o_$\mathcal{L}_Y$. (c) Results of GeAE and GeAE_w/o_$\mathcal{L}_R$.

Fig. 6. Ablation studies on different datasets.

in model efficiency due to the increasing number of model parameters. Therefore, the balance between model performance and efficiency should be considered when setting the value of m.

To illustrate the contributions of each component in the objective function defined in (1) during the training of GeAE, we show the trajectory of each component in Fig. 5. It indicates that the value of $\mathcal{L}_C$ remains relatively small compared to the other components, which may explain the model's reduced sensitivity to the variations in λ_c. However, the values of each component are of comparable magnitudes, justifying the choice of $\lambda_c = \lambda_y = \lambda_r = 1$.

E. Ablation Study

To examine the impact of different components of the GeAE objective function, we perform ablation studies on various datasets. The tasks are selected in the same way as in the parameter sensitivity analysis. To assess the effect of the causal structure learning loss $\mathcal{L}_C$, we remove $\mathcal{L}_C$ from the objective function while keeping the other components unchanged, referring to this version of GeAE as GeAE_w/o_$\mathcal{L}_C$. Similarly, to investigate the effect of the cross-entropy loss $\mathcal{L}_Y$ (and the regularization term $\mathcal{L}_R$), we use GeAE_w/o_$\mathcal{L}_Y$ (GeAE_w/o_$\mathcal{L}_R$, respectively), wherein $\mathcal{L}_Y$ ($\mathcal{L}_R$, respectively) is removed and the other components remain unchanged. The results are summarized in Fig. 6. The figures clearly reveal that $\mathcal{L}_C$, $\mathcal{L}_Y$ and $\mathcal{L}_R$ each exert a positive influence on the process of causal representation learning, highlighting their individual importance in enhancing the model's performance.

As we can see from the equations of $\mathcal{L}_C$ and $\mathcal{L}_Y$, they are both associated with $\boldsymbol{A}$. This indicates that the causal structure learning loss and the cross-entropy loss collaboratively contribute to the process of causal representation learning. To highlight the role of causal representation learning, we design a new ablation study. We let $\lambda_c = 0$ and fix $\boldsymbol{A}$ to be an identity matrix. In the cross-entropy loss, we replace the Markov blanket of the target label in the cross-entropy loss with the low-dimensional representation derived from the autoencoder. This method can be seen as a classical autoencoder for the classification task and is referred to as GeAE_w/o_Cau. The comparison results between GeAE and GeAE_w/o_Cau on different datasets are presented in Tables A6–A10 in Appendix, available online. A performance decline in GeAE_w/o_Cau is observed in most of the cross-domain classification tasks, underscoring the critical role of causal representation mechanism incorporated into the autoencoder model.

V. Conclusion

In this paper, we have introduced a new causal representation model, named GeAE, based on a graph autoencoder embedded autoencoder. GeAE performs causal structure learning in a modular similar to a graph autoencoder. It can efficiently extract nonlinear causal relationships included in the data only using one single source domain. Moreover, GeAE effectively combines the causal structure learning loss and the cross-entropy loss within a deep autoencoder framework. These characteristics make the obtained feature representation suitable for cross-domain classification tasks with completely unknown target data, and enhance the applicability of the method to high-dimensional cases. The experiments on synthetic and real-world datasets have demonstrated its effectiveness.

As we can see, the objective function of GeAE consists of four components. The choice of balancing parameters and network architecture significantly influences its performance. Determining the optimal balancing parameters and network architecture automatically is regarded as one of the pivotal research topics in the field of Automated Machine Learning (AutoML). In the future, we will investigate how to combine GeAE with AutoML to enable automatic causal representation learning for cross-domain classification tasks.

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